



MA 460: Nonlinear Dynamics and Chaos

Course Code	MA 460
Course Name	Nonlinear Dynamics and Chaos
L-T-P-C	3-0-0-3
Prerequisites	MA 101 and MA 102
Intended for	B Tech 3rd year
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Module I** [3 Lectures]
Introduction to Nonlinear Dynamics and Chaos, Recent applications of Chaos, Computer and Chaos, Dynamical view of the world.
- **Basics of nonlinear science** [5 Lectures]
Dynamics, Dynamical Systems, Types of Dynamical Systems, Nonlinearity, Dissipative Systems, Deterministic versus Stochastic Systems, Degree of Freedom, State Space, Phase Space, Attractor.
- **Module III** [4 Lectures]
Stability of solutions to Ordinary Differential Equations
- **Module IV** [5 Lectures]
Flows on line, Fixed Points and its Stability, Analytical Approach, Graphical approach, Simulation of Equations
- **Elementary Bifurcation Theory** [5 Lectures]
Saddle Node, Transcritical, Pitchfork, Imperfect, Hopf bifurcation
- **Module VI** [4 Lectures]
Two dimensional Flows, Simple Harmonic Mass-Spring Oscillator
- **Module VII** [4 Lectures]
Limit Cycle, Ruling out closed orbits, Poincare Benedixson theorem
- **Module VIII** [6 Lectures]
Butterfly Effect, Chaos, Lorenz Equations, Application of Chaos in sending secret messages, Introduction to Fractals, Dimensions of fractals, Cantor Set and Koch curve
- **Module IX** [5 Lectures]
One dimensional map, Logistic Map, Period doubling Route to chaos, Feigenbaum constants

Text Books

1. K.Allgood, T.Sauer, J.A.Yorke, **Chaos: An Introduction to Dynamical systems**, Springer Verlag
2. H.G. Solari, M.A. Natiello and G.B. Mindlin, **Nonlinear Dynamics: a two way trip from Physics to Maths**, Overseas publication

References

1. Ian Stewart, **Does God Play a Dice? The Mathematics of Chaos**, Blackwell, NewYork.
2. Laksmanan M Rajsekhar, **Nonlinear Dynamics Integrability Chaos and Pattern**, Springer.
3. F.C. Moon, **Chaotic and Fractal Dynamics**, Wiley.